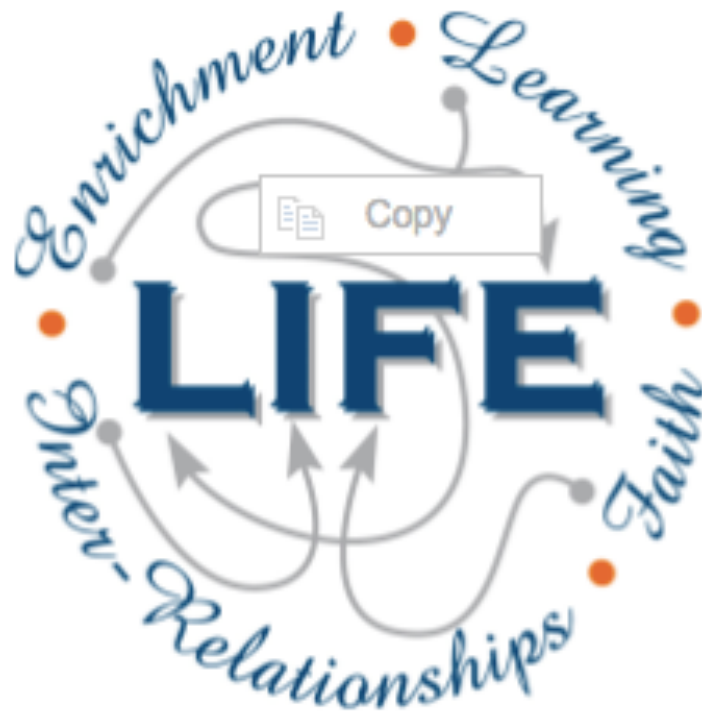




Year 10 Science

Earth & Space Sciences

2017/18 Programme

Programme

Week	Topic	Content	Activities and Assessment Tasks	Resources
1	Introduction to Astronomy	- Understand the expectations and objectives of the course. - Examine celestial objects, their relative size and scale, and how they fit into the cosmic scheme of things. - Describe the three different types of galaxies. - Understand that constellations are patterns formed by stars.	ScienceWeb - Lesson 1 (What's in our universe?)	JacPlus Ch 6.1
2	Size of The Universe	- Appreciate the vast distances involved in space travel. - Understand the unit light year, and perform simple calculations using it. Extension - Describe how the parallax method is used to determine stellar distances.	ScienceWeb - Lesson 2 (Size + distance in space.)	JacPlus Ch 6.1
3	Origins of the Universe	- Describe the competing theories on the origin of the universe - Creation, Steady State, Big Bang and Big Crunch. - Identify the evidence that supports each theory. - Understand that the Big Bang Theory is considered the best theory at the moment. - Understand that our universe is considered to be expanding. - Know the current distance to the edge of the "known" universe. - Understand how the Doppler Effect gives an indication of the movement of galaxies.	ScienceWeb - Lesson 3 (The Big Bang Theory.) Invest. 6.5 p. 205 SQ10	JacPlus Ch 6.3, 6.4
4			Research assignment: Origin of the Universe. (50%)	
5	Life and Death of Stars	- Describe the life of a star from birth to death, dependent on its size. - Understand the difference between a nova and supernova. - Describe the characteristics of a white dwarf, neutron star and black hole. - Describe the three possible pathways for the life of a star. Extension - Hertzsprung-Russell diagram for life cycles.	ScienceWeb - Lesson 4 (Life + death of stars.)	JacPlus Ch 6.2
6			Webquest: Life Cycles	
7	Observing the Universe	- Describe how radio telescopes and arrays of radio telescopes are used by astronomers and astrophysicists to observe distant parts of the universe. - Explain how orbiting space telescopes are used to gather data from deep space and how they compare with Earth-based telescopes. - Appreciate that the study of the universe and the exploration of space involves teams of specialists from different branches of science, engineering and technology.	ScienceWeb - Lesson 5 (Observing the universe.)	JacPlus Ch 6.5
8	Revision and Test		Topic Test (50%)	

Assessment Structure

Topic	Weighting	Assessment Type	Title and Details	Approximate Date
Earth & Space Sciences	10%	Research	Origin of the Universe	Week 6
	10%	Test	Topic Test	Week 8